

Palladium-Catalyzed C-H Activation and Arylation of Heteroarenes: Optimization and Scope

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Abstract:

Direct C–H bond functionalization provides an atom-economical methodology for constructing biaryl architectures. In this study, we investigate a phosphine-free palladium catalytic system for the regioselective direct C-H arylation of thiazole and benzoxazole cores. Under optimized catalytic conditions, Compound 5c was isolated in 88% yield with high functional group compatibility. The catalytic protocol operates under benchtop conditions at 90 °C in DMF without requiring strictly anhydrous precautions. The methodology was extended to fifteen distinct heterocyclic derivatives, establishing broad scope for late-stage pharmaceutical elaboration.

Introduction:

Transition-metal-catalyzed cross-couplings conventionally rely on pre-functionalized organometallic reagents such as stannanes or boronic acids. Direct C-H activation eliminates stoichiometric organometallic byproduct generation. In particular, tailoring catalyst loadings and ligand coordination environments remains pivotal for achieving high conversions without homocoupling side-products.

Results and Discussion:

Initial screening focused on coupling 2-phenylthiazole with 4-bromotoluene using various palladium precursors and carboxylate ligands. Pivalic acid (PivOH, 30 mol %) proved essential as a proton-shuttle additive to facilitate concerted metalation-deprotonation (CMD).

Table 1: Optimization of Reaction Conditions for Arylation

Entry	Catalyst (mol %)	Base (equiv)	Solvent	Temperature (°C)	Isolated Yield (%)
1	Pd(OAc) ₂ (2.5)	K ₂ CO ₃ (2.0)	DMF	90	71%
2	Pd(OAc) ₂ (5.0)	Cs ₂ CO ₃ (2.0)	Toluene	110	54%
3	PdCl ₂ (MeCN) ₂ (2.5)	K ₂ CO ₃ (2.0)	DMF	90	63%
4	Pd(OAc) ₂ (2.5)	KOAc (2.0)	NMP	100	66%

Table 1 demonstrates that Entry 1 provided the optimal balance of efficiency and selectivity, affording Compound 5c in 71% isolated yield. Reaction temperatures below 80 °C led to incomplete starting material consumption, while temperatures exceeding 110 °C caused visible palladium black precipitation and substrate degradation.

Substrate Scope and Derivative Analysis:

With optimized conditions in hand (Pd(OAc)₂ 2.5 mol %, PivOH 30 mol %, K₂CO₃ 2.0 equiv, DMF, 90 °C, 12 h), the scope of aryl halides was examined. Both electron-rich and electron-poor aryl bromides delivered functionalized adducts in good to moderate yields.

Table 2: Isolated Yields of Biaryl Adducts

Derivative	Aryl Substituent (R)	Isolated Yield (%)	Melting Point (°C)
Compound 5a	4-OMe	78%	112–114
Compound 5b	4-CF ₃	65%	98–100
Compound 5c	4-Me	71%	104–106
Compound 5d	3-CN	59%	145–147

As detailed in Table 2, electron-donating substituents slightly enhanced product formation (Compound 5a, 78% yield). For Compound 5c bearing a 4-methyl substituent, an isolated yield of 71% was obtained after column chromatography.

Experimental Procedures:

General Procedure for Direct C-H Arylation:

A 25 mL resealable Schlenk tube equipped with a magnetic stir bar was charged with 2-phenylthiazole (1.0 mmol), aryl bromide (1.2 mmol), Pd(OAc)₂ (5.6 mg, 0.025 mmol, 2.5 mol %), pivalic acid (30.6 mg, 0.30 mmol, 30 mol %), and potassium carbonate (276 mg, 2.0 mmol, 2.0 equiv). Anhydrous DMF (3.0 mL) was added. The tube was sealed and stirred in a preheated oil bath at 90 °C for 12 h. Upon completion, the mixture was cooled to room temperature, diluted with ethyl acetate (25 mL), washed with brine (3 × 15 mL), dried over anhydrous Na₂SO₄, and concentrated in vacuo. The crude residue was purified by flash column chromatography on silica gel (hexane/EtOAc 10:1).

Synthesis of 5-(p-tolyl)-2-phenylthiazole (Compound 5c):

Pale yellow crystalline solid; mp 104–106 °C; isolated yield: 71% (178 mg, 0.71 mmol).

Note: Crude ¹H NMR analysis of the unpurified reaction mixture indicated an 88% NMR conversion, whereas chromatographic isolation furnished the purified product in 71% isolated yield.

¹H NMR (400 MHz, CDCl₃): δ 8.08 (s, 1H), 7.96–7.92 (m, 2H), 7.54 (d, J = 8.0 Hz, 2H), 7.45–7.38 (m, 3H), 7.23 (d, J = 8.0 Hz, 2H), 2.38 (s, 3H).

¹³C NMR (100 MHz, CDCl₃): δ 166.4, 142.0, 139.1, 138.4, 133.6, 129.8, 129.6, 128.9, 126.8, 126.3, 21.3.

HRMS (ESI) m/z calculated for C₁₆H₁₄NS [M+H]⁺ 252.0847, found 252.0842.